Performance Assessment of Heaving Point-Absorber Wave Energy Converters at Harbour Locations in YYYY

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ABSTRACT**:** This paper presents a preliminary assessment of the performance of heaving point-absorber wave energy converters deployed at selected harbour locations in YYYY. Long-term offshore buoy time-series were processed and analysed using univariate, bivariate, and directional statistical methods to characterize the local wave climate and its seasonal variability. The device response under both regular and irregular wave conditions was evaluated using two power take-off (PTO) formulations: a generic linear damping model and a simplified hydraulic representation. The influence of the mooring system was incorporated through both quasi-static and dynamic approaches, considering two common configurations: a single-line mooring and a three-leg arrangement. Energy conversion metrics were computed for selected years at the deployment site, and a site-specific optimization of the conversion system was performed to identify performance improvements under local wave conditions.

Keywords: Heave point absorber, PTO non-linear force, site optimization

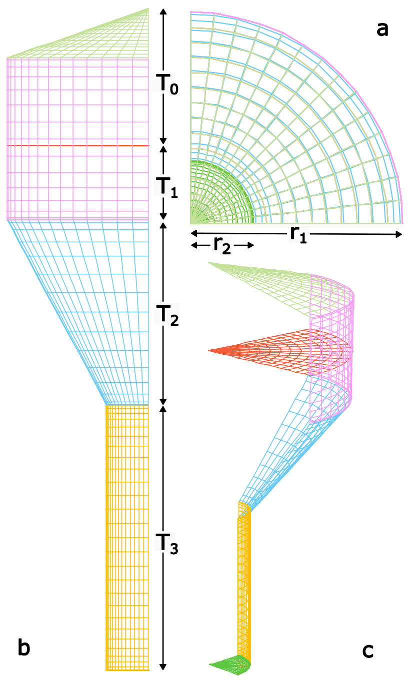

Figure - C4-Like (a) Top (b) Side (c) 3D views

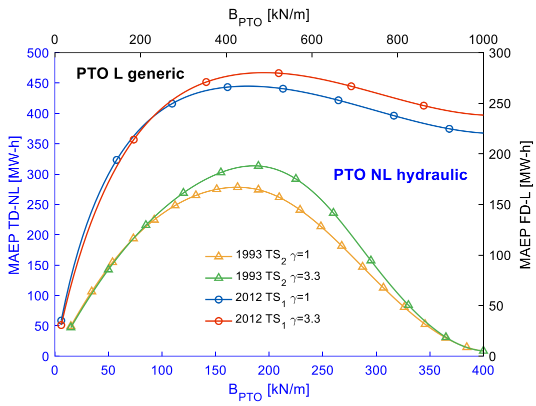

Figure – MAEP for non-linear model and linear model comparison